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SOSexp2.m | SOSexp2c.m

```
1   % Experiment 2(iii): Addition and Multiplication of Two Sequences
2   clc; clear; close all;
3   n = 0:2;
4   x = [1 2 1];
5   y = [1 1 2];
6   zSum = x + y;
7   zProd = x .* y;
8   figure;
9   subplot(2,2,1); stem(n,x,'filled'); grid on;
10  title('x[n]'); xlabel('n'); ylabel('Amplitude');
11  subplot(2,2,2); stem(n,y,'filled'); grid on;
12  title('y[n]'); xlabel('n'); ylabel('Amplitude');
13  subplot(2,2,3); stem(n,zSum,'filled'); grid on;
14  title('Addition x[n]+y[n]'); xlabel('n'); ylabel('Amplitude');
15  subplot(2,2,4); stem(n,zProd,'filled'); grid on;
16  title('Multiplication x[n]y[n]'); xlabel('n'); ylabel('Amplitude');
```

Command Window

Workspace

Name	Value
n	[0,1,2]
x	[1,2,1]
y	[1,1,2]

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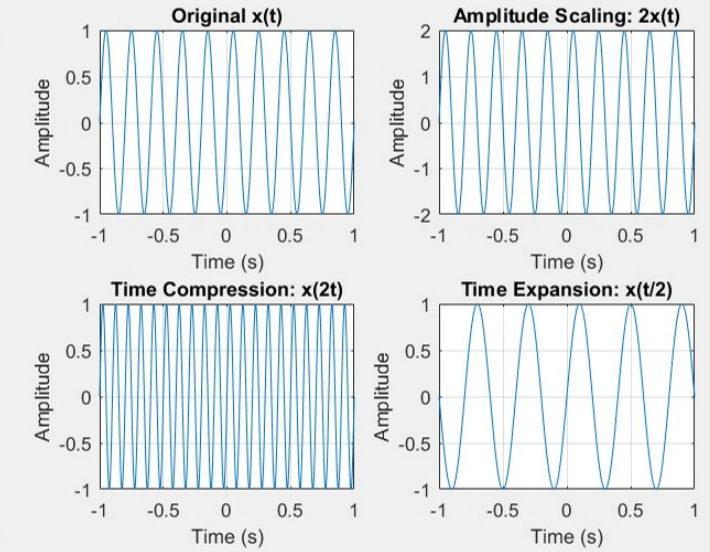
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SOSexp2.m

```
1   % Experiment 2(i): Amplitude and Time Scaling
2   clc; clear; close all;
3   t = -1:0.001:1;
4   x = sin(2*pi*5*t);
5   yA = 2*x;
6   yC = sin(2*pi*5*(2*t));
7   yE = sin(2*pi*5*(t/2));
8   figure;
9   subplot(2,2,1); plot(t,x); grid on;
10  title('Original x(t)'); xlabel('Time (s)'); ylabel('Amplitude');
11  subplot(2,2,2); plot(t,yA); grid on;
12  title('Amplitude Scaling: 2x(t)'); xlabel('Time (s)'); ylabel('Amplitude');
13  subplot(2,2,3); plot(t,yC); grid on;
14  title('Time Compression: x(2t)'); xlabel('Time (s)'); ylabel('Amplitude');
15  subplot(2,2,4); plot(t,yE); grid on;
16  title('Time Expansion: x(t/2)'); xlabel('Time (s)'); ylabel('Amplitude');
```

Figure 1

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Workspace

Name	Value
t	1x2001 double
x	1x2001 double
yA	1x2001 double

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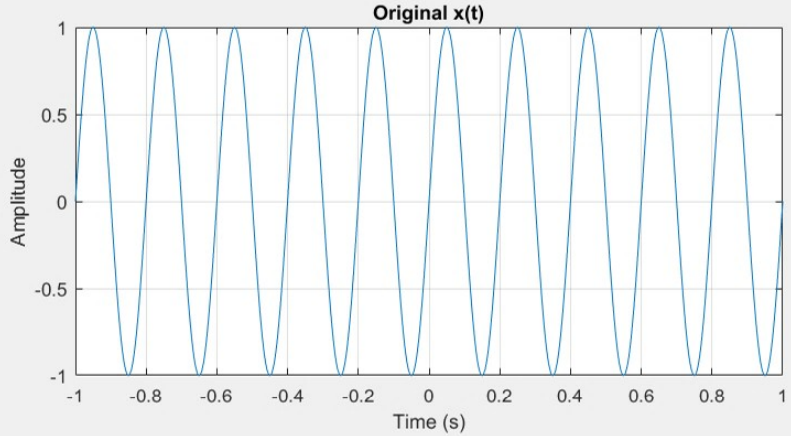
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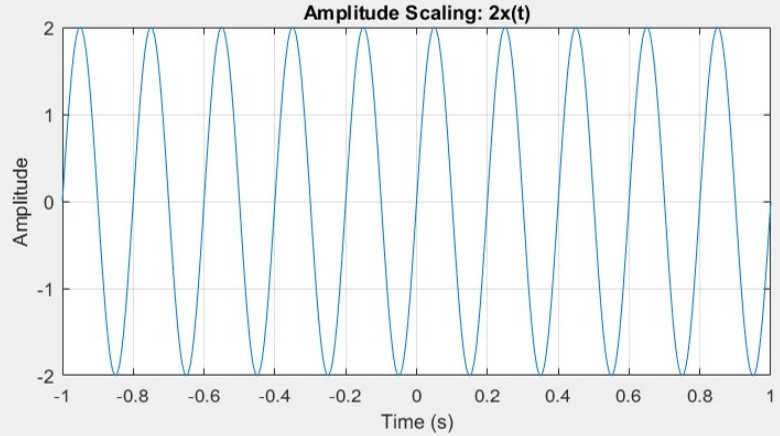
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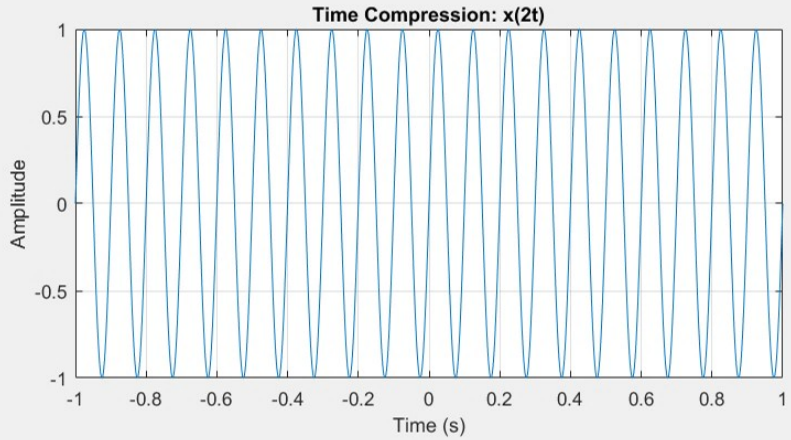
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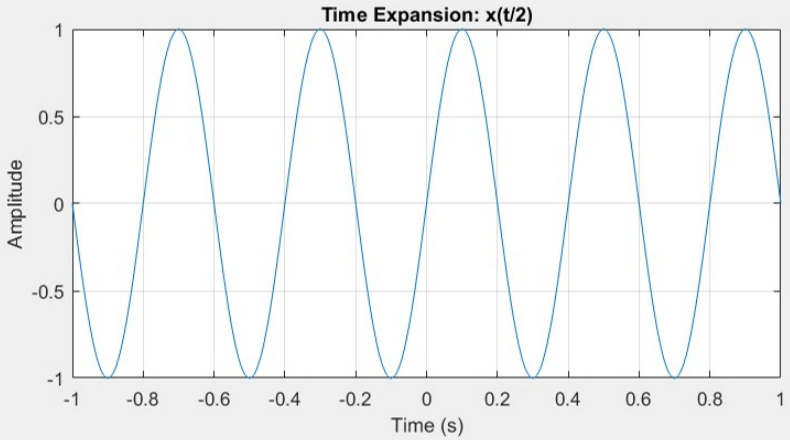
Figure 1

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Command Window

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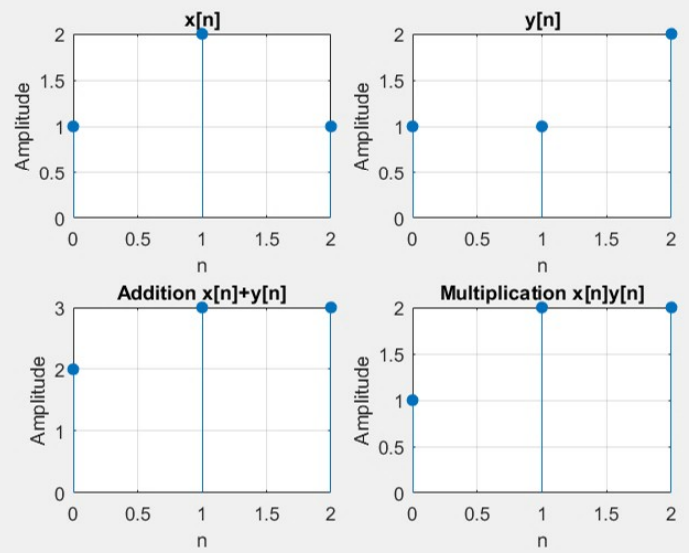
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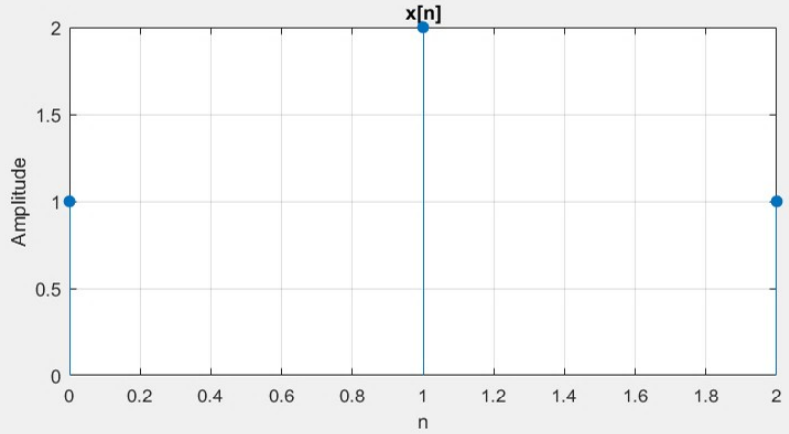
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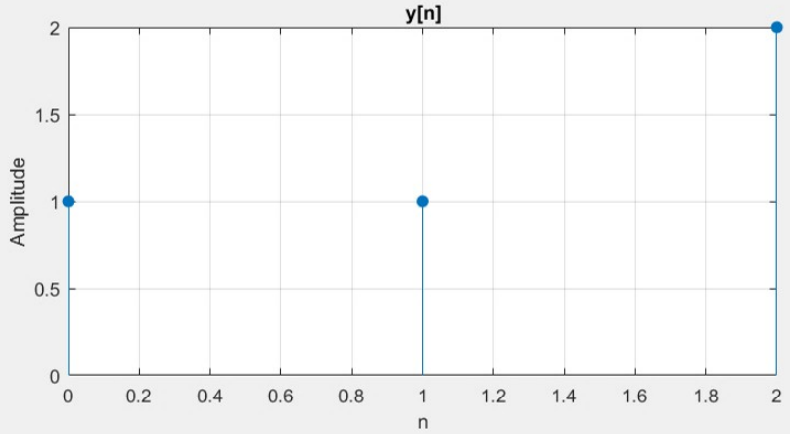
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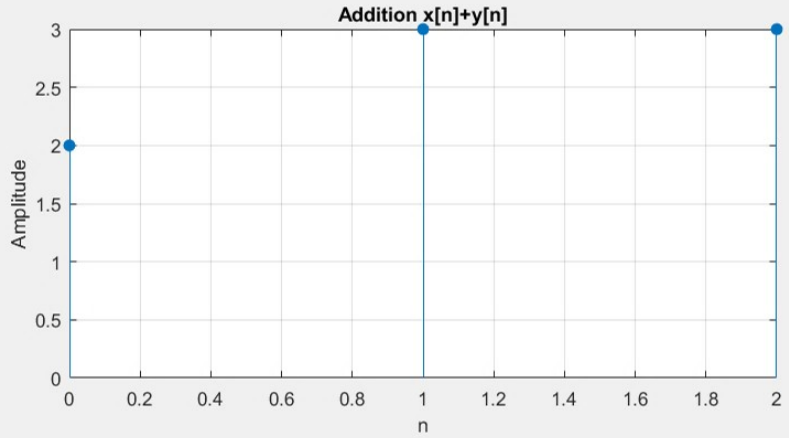
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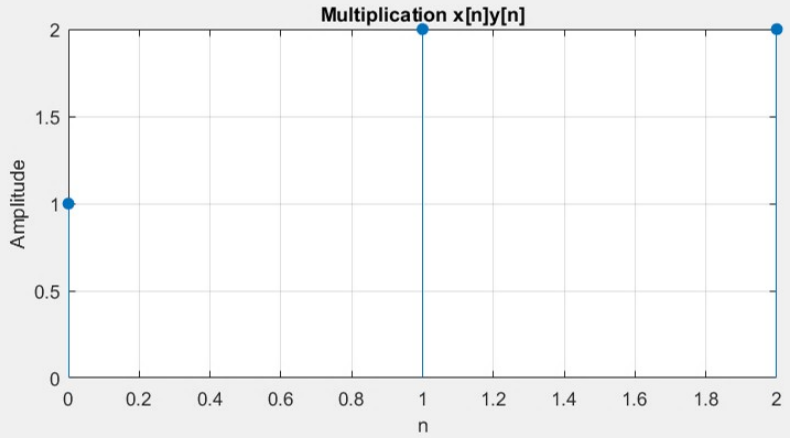
Figure 1

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